Short-Horizon Forecasting of the SPY Implied-Volatility Surface

A pre-registered study on Databento OPRA data

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Abstract

Are short-horizon moves of the SPY implied-volatility surface predictable, and does the forecast retain positive vega-attributed edge after observable spread costs? I build one-minute, quality-gated and static-arbitrage-screened volatility-surface states from OPRA consolidated top-of-book quotes (4.4×10^{10} records over the initial 21-session panel, June 2–July 1, 2026, later extended to 64 sessions spanning March 31–July 1), then forecast per-cell implied-volatility changes at 5–60-minute horizons under a purged walk-forward protocol with mechanical leakage probes and shuffled-label negative controls. At the primary 15-minute horizon, ridge regression on a compact set of time-t features improves vega-weighted MSE over the persistence null by 16.7% on the June panel (25.9% unweighted); on the expanded three-month panel its unweighted skill is 26.1%. A GPU sweep of temporal convolutional networks shows a deep sequence model adds essentially nothing: within the tested configurations, receptive fields beyond 15 minutes produced no additional out-of-sample skill. Pre-registered feature ablations reveal that a finding as intuitive as "Greeks help" flips sign between a one-month and a three-month panel. A final symmetric experiment finds that, in the tested one-minute per-cell representation, Lee–Ready-signed prints add no measurable incremental out-of-sample skill beyond quote-based order-flow features at any horizon. Every expectation was registered before the corresponding run; negative results are reported as registered. The economic question is assigned to a pre-registered execution-aware replay (Section 6) whose design, endpoints, and verdict taxonomy were locked before any replay computation.

1 Question and setup

The object being forecast is the surface state $\mathrm{IV}_t(\text{cell})$: for each cell of a fixed (days-to-expiry $\times$ log-moneyness) grid, the target is the forward change

$$\Delta\mathrm{IV}(t,h) \;=\; \mathrm{IV}(t+h) - \mathrm{IV}(t), \qquad h \in \{5, 15, 30, 60\} \text{ minutes},$$

with 15 minutes as the primary horizon. The null model is persistence ($\Delta\mathrm{IV} = 0$) — the surface stays where it is — which at these horizons is a genuinely hard baseline, not a strawman.

"Good" is defined by what a desk would do with the forecast, not by raw RMSE. The eventual use is relative-value: to first order, the volatility-attributed premium change of an option is approximately $\mathrm{vega}_t \times \Delta\mathrm{IV}$ per share (dollar results per standard contract multiply quoted premium changes by the 100-share multiplier). This first-order attribution excludes delta, gamma, theta, and volga effects, hedging, and financing; the execution replay of Section 6 therefore reports a *vega-linearized*

*Forecasting results correspond to `qfdev_research` commit `3eba5bf8`; the execution-replay design of Section 6 is locked at commit `613a309f`, pushed before any replay computation.

edge, never total option P&L. The headline forecasting metric is *vega-weighted* MSE skill versus persistence (defined exactly in Section 3), alongside the per-minute cross-sectional rank information coefficient. The tradeable scope was locked in advance to short-dated wings — expiries within seven days, strikes off the immediate at-the-money point — because that is where both the movement and a realistic edge concentrate.

2 Data and surface construction

The quote source is Databento OPRA consolidated NBBO top-of-book (`cmbp-1`) for all listed SPY options, with Nasdaq `mbp-1` for the underlying. Three nested samples appear in this memo:

Stage	Dates (2026)	Sessions	Purpose
Initial panel	Jun 2 – Jul 1	21	Baselines and first ablations
Expanded panel	Mar 31 – Jul 1	64	Architecture sweep; pre-registered replications
Replay prediction dump	Apr 15 – Jul 1	54 test days	Execution-aware replay (Sec. 6)

Table 1: The three sample stages. The replay dump contains the 7,633,773 out-of-sample prediction rows produced by the walk-forward protocol on the expanded panel; its 54 test days are the expanded panel's out-of-sample days.

The initial 21-session panel (June 2 to July 1; "the June panel" below) covers roughly 858 GB of raw quotes, 44.05 billion records, verified row-for-row against the vendor's manifest.

Quotes are compacted into one-minute surface snapshots. Each snapshot fits SVI slices [2] per expiry with quality gating (85.7% of slices trusted on the reference validation day; the rest fall back to direct interpolation), recovers forwards from put–call parity (100% coverage), and writes IV, Black-76 [1] Greeks, and quality flags onto a fixed grid of 9 DTE buckets $\times$ 14 log-moneyness nodes — 126 cells per minute. The June panel used for forecasting is 886,467 cell-minutes.

Every session must pass an automated validation gate before it enters the panel. Among contract-minutes passing the quote-eligibility filters (finite two-sided quotes with $0 \leq \text{bid} \leq \text{ask}$ and a positive mid — the builder's `ok` flag), the fraction with a solvable IV must be $\geq 90\%$; coverage fractions quoted against *all* raw contract-minutes, including ineligible and malformed quotes, are lower by construction and are not the gated quantity. Further gates: median put–call IV disagreement ≤ 1.5 vol-points, butterfly-negativity (a static-arbitrage smell) below threshold, SVI fit RMSE bounds, and an external sanity check of ATM IV against an independent reference within 3 vol-points. All 21 June sessions passed.

3 Methodology and pre-registration

Models. Five model classes appear throughout: persistence (the $\Delta\text{IV} = 0$ null); a constant train-mean baseline (the drift-aware null used by the negative controls); ridge regression ($\alpha = 1$ on standardized features) as the locked reference model; LightGBM as a vega-weighted challenger; and a causal dilated temporal convolutional network (TCN) [4] trained with a vega-weighted Huber loss.

Vega-weighted metric. The headline metric is skill versus persistence under vega-weighted MSE,

$$\text{MSE}_v = \frac{\sum_i w_i \, (y_i - \hat{y}_i)^2}{\sum_i w_i}, \qquad w_i = \text{vega}_i,$$

where vega_i is the raw per-cell Black-76 vega written by the surface builder at decision time t (no cap, floor, or normalization; rows with missing vega are excluded), and skill is $1-\text{MSE}_{\text{model}}/\text{MSE}_{\text{persistence}}$ computed under the same weights. Weighting always enters *evaluation*; it enters *training* only where stated — LightGBM via per-row sample weights and the TCN via its vega-weighted loss; ridge is trained unweighted.

Cross-sectional rank IC. At each minute with at least 20 in-scope cells, the Spearman rank correlation between predicted and realized ΔIV is computed across cells; the reported figure is the unweighted mean over minutes. Because 15-minute labels are sampled every minute, the raw t-statistics over minutes are overlap-inflated; deflating by $\sqrt{15}$ is quoted alongside where it matters.

Validation protocol. Five-fold expanding-window walk-forward over the minute axis, purged and embargoed by the longest horizon (60 minutes) so overlapping labels cannot leak across the train/test boundary [6]; roughly 600–700k out-of-sample rows per horizon on June alone. Feature standardization is fit on training folds only.

Leakage controls. Two mechanical probes run against the feature builder itself: truncation invariance (features computed on a panel cut at time t must be bitwise identical to the full-panel build) and future-input perturbation (corrupting all raw inputs after t must leave features at or before t unchanged). The probes were validated by injecting a deliberate future-IV feature, which they caught. In addition, shuffled-label negative controls were run for every model class: within each walk-forward fold, *training* labels are permuted after the purge/embargo (test labels intact) and the models retrained. Because the target has non-zero mean over long windows, shuffled models are judged against the constant train-mean baseline rather than the zero null — otherwise target drift masquerades as leakage. Measured feature-driven excess collapses to $\leq +0.004$ against a 0.02 tolerance (versus +0.17 to +0.31 on real labels), and feature importances go flat.

Significance tests. Reported Diebold–Mariano [5] p-values test the per-minute loss-differential series between two arms scored on identical row sets (cross-sectional losses aggregated within the metric's weighting), with a Newey–West long-run variance at lag equal to the horizon length in minutes to absorb label overlap. They are computed per (model, horizon, metric) cell and are *not* adjusted for multiplicity across the grid; they serve as supporting diagnostics. The primary evidence standard everywhere is the registered effect threshold with the sign holding in every walk-forward fold and every calendar month.

Pre-registration. Before each experiment round, the expected outcome, the primary endpoints, and the decision rules were written down and committed. Example: Greeks become default-on for a model class only if they add at least +1.0 percentage point of vega-weighted skill on at least two horizons *with the sign holding in every walk-forward fold.* The point is to make the negative results as legible as the positive ones.

4 Results

4.1 Baselines beat persistence; the signal is mostly linear

Walk-forward skill versus persistence on the June panel:

Skill vs. persistence	5m	15m	30m	60m
Ridge, plain MSE	+25.2%	+25.9%	+24.6%	+30.9%
LightGBM, plain MSE	+21.5%	+16.7%	+18.3%	+24.9%
Ridge, vega-weighted	+19.6%	+16.7%	+8.3%	+8.0%
LightGBM, vega-weighted	+17.8%	+11.9%	+7.0%	+1.3%

Table 2: Out-of-sample skill, 21-session June panel, 5-fold purged walk-forward. LightGBM rows use vega sample-weighting in training. Values are point estimates; day-clustered uncertainty is currently reported only for the registered adjudications (fold ranges, DM diagnostics) and the replay's day bootstrap — extending day-bootstrap intervals to this table is queued work (Section 7).

Ridge matches or beats LightGBM on plain MSE at every horizon: the signal is largely linear in this feature set. The weighting matters more than the model: an unweighted LightGBM is *worse than persistence* at the 15-minute horizon once evaluated vega-weighted (-1.3%), and simply training with vega sample-weights swings it to +11.9% — a 13-point difference from an evaluation choice, before any modeling work. Directional accuracy decays from 58% at 5 minutes to 53% at 60 minutes. On short-DTE wings, the locked reference model achieves a mean per-minute cross-sectional rank IC of +0.40 (raw $t \approx 116$ over minutes, ≈ 30 after $\sqrt{15}$ overlap deflation; 91% of minutes positive) with a vega-weighted calibration slope of 0.992 (vega-WLS of realized on predicted through the origin) — predicted magnitudes can be taken at face value in the tradeable scope. The adjacency of surface cells makes the per-minute cross-section dependent, so the IC's minute count overstates its effective sample size; the fold- and month-consistency standards of Section 3 are the load-bearing evidence.

4.2 A negative result on architecture

An 11-run TCN sweep on the full 64-session span (single RTX 4060 Ti, ∼15h wall) tested whether a deep sequence model finds anything the cheap models miss. It does not: TCN skill ties ridge and LightGBM at every horizon — at 15 minutes the *plain-MSE* skill fractions are ridge 0.261, LightGBM 0.251, TCN 0.248 (the sweep's summary metric is unweighted skill versus persistence). A receptive-field ablation is the sharpest summary: growing the TCN's context from 3 to 7 minutes helps, and within the tested configurations, increasing the receptive field beyond 15 minutes produced no additional out-of-sample skill. That bounds the incremental value of longer memory for this model class and representation; it does not establish that no longer-memory information exists. In a capacity sweep, additional dropout did not improve out-of-sample performance (dropout 0 was optimal), suggesting insufficient regularization was not the limiting factor in the tested capacity range. The feature set does the work, not the architecture; the next lever is cross-sectional structure, not a bigger network.

4.3 Feature ablations: importance is not generalization

On June, adding Black-76 Greeks as features *hurt* LightGBM by 4 to 11 skill points depending on horizon (worst at 30m: -10.6pp), while leaving ridge indifferent — even though Greeks captured 8.6% of the model's split gain at that same horizon. A plausible interpretation was that gamma, theta and vega are near-deterministic transforms of features already present, and that the redundancy encouraged unstable tree splits that spend themselves on out-of-sample variance. The expanded-panel replication in Section 5 was designed to test exactly that interpretation. Quote-based microstructure features (order-flow imbalance, quote imbalance) added +1.5pp for LightGBM at 30/60m and nothing below that. Both findings were treated as hypotheses to re-test, not conclusions.

5 What changed between rounds — and why

The June ablations were re-run as a pre-registered 2×2 (Greeks $\times$ microstructure) on the three-month panel — the five model classes of Section 3 $\times$ 4 horizons $\times$ 4 arms — with predictions and decision rules committed before launch. The registered expectations were: ridge flat under Greeks ($|\Delta| < 0.5$pp), LightGBM hurt by ≥ 1pp on ≥ 2 horizons (replicating June), TCN approximately flat, and no Greeks$\times$micro interaction.

The replication broke the June story in an instructive way. All contrasts below are computed on identical row sets, with per-fold and per-calendar-month deltas and Diebold–Mariano tests as described in Section 3. The LightGBM Greeks harm — the largest and most intuitive June effect — did *not* replicate: at the 30-minute replication target the sign flipped from -10.6pp to $+0.7$pp (all five folds positive, DM $p < 10^{-4}$), with $+0.3$ to $+0.7$pp across the other horizons. Ridge, registered as flat, instead gained $+1.9$pp of vega-weighted skill at 30 minutes and $+4.8$pp at 60 from Greeks, with the sign holding in every walk-forward fold and every calendar month — so under the rule fixed in advance, Greeks flipped default-on for the ridge class (and only there: no other model class met the threshold; a fold-consistent TCN cell at 30 minutes had no vega-weighted counterpart and was discarded under the registered stability standard). The registered mechanism story — that the June harm was specific to greedy split selection — failed its own preconditions and is reported as wrong. The microstructure edge shrank from $+1.5$pp to $\sim$+0.3pp at 30/60 minutes; its registered replication target (LightGBM 60m $\Delta > 0$) formally held ($+0.31$pp, all folds positive, DM $p = 0.026$) but attenuated fivefold. At the 15-minute story horizon every effect was null. The plain reading: one month is one regime, and feature-ablation signs measured on it do not transfer. The pre-registration is what makes this a finding rather than an embarrassment — the June conclusions were recorded, the re-test criteria were fixed in advance, and the reversal is reportable exactly because nothing was re-fit after seeing the answer.

A symmetric test of trade flow

The quote-based order-flow features raise an obvious question: do actual *prints* carry anything the book does not? A second pre-registered 2×2 (microstructure $\times$ signed trade flow) added Lee–Ready [3] signed volume per cell from OPRA trades, engineered with exactly the same 30-minute temporal treatment as the quote-pressure features so the comparison could not be confounded by representation, and framed two competing hypotheses in advance: prints add information beyond posted pressure (H-new) versus quotes impound trades within the minute (H-span). Results consistently favored H-span over H-new within the registered representation. The primary endpoint — LightGBM skill change from flow given the quote features, 60-minute horizon — was -0.07pp (DM $p = 0.46$); the largest ridge cell was $+0.07$pp; and the substitution contrast (flow alone versus neither) was equally null, so there was no flow credit for the quote features to steal. Two TCN cells produced large point estimates ($+3.0$pp at 30m vega-weighted, DM $p = 0.004$) with per-fold deltas spanning -1.1 to $+11.5$pp — exactly the multiple-comparisons bait the registered fold-consistency standard exists to discard. Two caveats are owned rather than hidden: Lee–Ready misclassifies option trades (multi-leg packages especially), and per-cell signed volume conflates directional with volatility demand — the strongest untested variant, chain-level vega-weighted net flow, is named in Section 7 rather than quietly run.

The replication round also doubled as an engineering exhibit: the two overlapping arms were recomputed on a second machine (Windows, pinned to the same library versions) and all 64 recomputed ridge and LightGBM summary values matched the Linux results bit-for-bit. (The claim is scoped to those table values; row-level prediction files were not hashed.)

6 Execution-aware replay: does the edge survive the spread?

Vega-weighted forecast skill is not a trading result. This section closes that gap with a pre-registered execution replay that maps each in-scope cell forecast to an actual quoted contract and charges observable spread costs. The full design, endpoints, falsifiable expectation bands, and verdict taxonomy below were committed (`613a309f`) *before any replay computation existed*; the replay runs exactly once against them.

6.1 Registered design (summary)

For each (minute, cell) a single representative OTM-side contract is chosen on time-t information only (nearest log-moneyness to the node among quote-valid contracts, tie-broken by relative spread); the exit is the *same* contract at the exact quote h minutes later, with no forward fill — a missing exit is counted as unfillable and excluded from markout, never silently filled. Vega is the per-contract Black-76 vega at t. A trade fires iff $|\text{vega}_t \cdot \lambda \cdot \widehat{\Delta\text{IV}}| > \theta \cdot \text{half-spread}_t + s$, with $\theta = 1$, $\lambda = 1$, $s = 0$ primary and $\theta \in \{1, 1.5, 2\}$, slippage $s \in \{0, 0.005, 0.01\}$ \$/share, and a walk-forward-calibrated λ arm as sensitivities. Three outcome quantities are kept distinct: the gross vega-linearized edge $\text{sign}(\widehat{\Delta\text{IV}}) \cdot \text{vega}_t \cdot (\text{IV}_{t+h} - \text{IV}_t)$; the *entry-cost-adjusted markout* (gross minus half-spread at entry, marked to $t{+}h$ mid) — the statistical primary; and the *round-trip-touch net* (additionally minus the exit half-spread) — which drives the economic verdict. Scope is the registered tradeable region (DTE ≤ 7, off-ATM); out-of-scope cells are evaluated hypothetically and reported separately. Inference is a day-level bootstrap ($N = 10{,}000$) over the 54 test days. A greedy per-cell cooldown variant provides a non-overlapping replay; its cumulative output is a cumulative one-contract non-overlapping edge, *not* a portfolio equity curve, and is reported with concurrency diagnostics. A delayed-entry arm (decide at t, enter at the $t{+}1$-minute touch) bounds the latency objection.

6.2 Primary endpoint and decision rule

> *The registered primary endpoint is the pooled mean entry-cost-adjusted vega-linearized markout per fillable gated trade — ridge, 15-minute horizon, in-scope, $\theta = 1$, $\lambda = 1$, $s = 0$ — with a 95% day-bootstrap confidence interval.*

The endpoint is invalidated (reported as "diagnostics only", no headline claim) if gated exit unfillability exceeds 5% overall or is more than twice the overall rate in the top predicted-edge decile — selective unfillability where the model claims the most edge would make the markout uninterpretable.

6.3 Registered verdict taxonomy

The prose conclusion is constrained to one of three registered verdicts:

- **Survives observable entry costs.** Primary CI entirely above zero, round-trip-touch point estimate ≥ 0 with its sign preserved in the non-overlapping variant, validity condition intact.
- **Markout survives entry costs; round-trip execution not demonstrated.** Primary CI above zero, but the round-trip criterion fails.

- **Does not survive costs.** Primary CI straddling or below zero, or the endpoint invalidated by selective missing exits — reported as such: 15-minute surface forecasts of this quality do not clear the touch.

Six falsifiable expectation bands (gate pass rate, gross selection monotonicity in θ, the honest spread-survival band, LightGBM overconfidence as a trading statement, out-of-scope deterioration, and non-overlap sign preservation) were registered alongside; the strong result — round-trip positive with CI above zero — is registered as a *surprise*, not the expectation.

6.4 Results

[Pending. The replay had not been run when this draft was fixed; this section is populated from the registered run's output tables — primary endpoint with day-bootstrap CI, the θ/cost sensitivity grid, non-overlap replay with concurrency diagnostics, and the honesty diagnostics (unfillability profiles, linearization checks, delayed-entry arm) — and the verdict is stated in the registered taxonomy above.]

7 Limitations and next steps

- **One underlier, one quarter.** Everything here is SPY, March 31 – July 1, 2026 (approximately Q2). The regime-sensitivity result in Section 5 is itself the argument for not extrapolating further than the data supports.

- **Execution-aware markout, not a portfolio backtest.** The replay of Section 6 maps cell forecasts to actual option quotes and charges observable spread costs, but its primary outcome is a vega-linearized markout. It does not model delta hedging, nonlinear Greeks, queue position, displayed-size constraints, market impact, portfolio capital, or asynchronous latency (the delayed-entry arm bounds, but does not remove, the latency objection). The same-bar touch assumption is an execution benchmark, not a live-fill claim. Source quotes may have been forward-filled by up to five minutes in the surface builder, and the slim contract extract does not carry a per-row staleness flag.

- **Point estimates in the forecasting tables.** The headline skill tables report point estimates; day-clustered uncertainty currently exists only where the registered adjudications required it (fold ranges, per-month splits, DM diagnostics) and in the replay's day bootstrap. Extending day-bootstrap intervals to Table 2 is queued.

- **Surface gates are static-arb smells, not proofs.** The validator checks butterfly negativity and put–call consistency; a full calendar/butterfly no-arbitrage repair across the grid is open work.

- **The flow null is a null for one representation.** Per-cell, one-minute, Lee–Ready-signed volume with window aggregates adds nothing; that bounds the *average* contribution, not a localized one. The named untested variants: chain-level vega-weighted net flow (attacks the dealer-inventory channel directly, endpoint: LightGBM vega-weighted skill at 30/60m), skill conditioned on cell-minutes with nonzero window flow (attacks sparsity dilution), and multi-leg print exclusion (attacks sign misclassification).

- **Next:** run the registered replay exactly once and report against its bands; cross-sectional feature structure (the direction the architecture sweep points to); per-model feature defaults in code (the ridge/Greeks flip is adjudicated but ridge and LightGBM currently share one feature matrix).

References

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